

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Medical Diagnosis Center by Service Portal (ServiceNow)

Date	02 October 2026
Project Name	Medical Diagnosis Center by Service Portal (ServiceNow)
Maximum Marks	4 Marks

Functional Requirements

The following functional requirements define the core behaviour of the proposed ServiceNow-based Medical Diagnosis Center solution. They are adapted from the supplied Solution Requirements reference structure and aligned to the diagnostic booking, workflow, approval, record-update and notification scope established for this project. Capabilities not demonstrated in the current implementation are treated as planned or future scope rather than claimed as implemented.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Diagnostic Test Catalog	Provide a Service Catalog item for patients/users to submit diagnostic test requests. Acceptance: the configured request information is captured and a ServiceNow Requested Item (RITM) is created from the submission.
FR-2	Patient & Test Data Capture	Capture required patient and diagnostic information through catalog variables and mandatory fields. Acceptance: configured variables are available to the workflow and incomplete required information cannot be submitted.
FR-3	Dynamic Form Behaviour	Use clear form behaviour to show or require fields according to the selected diagnostic test or request context. Acceptance: conditional visibility and mandatory behaviour work as configured without unnecessary client-side logic.
FR-4	Test Booking & Workflow	Trigger the diagnostic booking workflow when the catalog request is submitted. Acceptance: the Service Catalog trigger receives the Requested Item record and starts the configured workflow.
FR-5	Appointment Variable Retrieval	Retrieve selected appointment and diagnostic variables for downstream automation. Acceptance: the configured test, date, time-slot and patient variables are exposed as workflow outputs.
FR-6	Diagnostic Booking Record Creation	Create a structured diagnostic booking/operational record from the submitted request values. Acceptance: configured catalog values are mapped into the operational record and the initial status is set to Requested.
FR-7	Approval Management	Send diagnostic requests requiring approval through the configured approval action. Acceptance: an approval request is generated and the configured approver/approval rule can provide an outcome.
FR-8	Approval Decision & Branching	Evaluate the approval outcome and execute the correct workflow path. Acceptance: approved requests follow the Approved branch; non-approved requests follow the corresponding rejection/else path and record update.
FR-9	Patient Notifications	Send lifecycle notifications at configured diagnostic-request stages. Acceptance: the request-created notification is sent after record creation and the configured status/resolution notification is sent after the relevant update.
FR-10	Appointment Status & Tracking	Maintain an understandable appointment/request status across the workflow. Acceptance: the operational record reflects configured lifecycle states such as Requested, Approved, Rejected, Completed or Report Available, where configured.
FR-11	Test Fulfilment & Report Handoff	Make approved diagnostic-request information available for downstream test fulfilment, sample collection, processing and report handoff. Acceptance: the approved request contains the mapped information needed by the responsible operational process; detailed fulfilment automation is a future extension unless configured.
FR-12	Workflow Monitoring & Documentation	Support workflow execution review, troubleshooting and solution handoff. Acceptance: administrators can inspect execution details, mappings and condition paths and maintain documentation for recreation in another ServiceNow instance.

Non-functional Requirements

The non-functional requirements define the quality expectations for usability, security, reliability, performance, availability, scalability, maintainability, data integrity, compatibility and auditability. These requirements are expressed at the application-design level; platform-level guarantees remain dependent on the deployed ServiceNow instance.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The patient-facing diagnostic booking experience should be simple, readable and guided. Mandatory fields and clear descriptions should reduce incomplete submissions.
NFR-2	Security	Access to patient, appointment and diagnostic data should be governed through ServiceNow roles and ACLs, with administrative configuration restricted to authorized users.
NFR-3	Reliability	For valid submissions, catalog data should move consistently from the Service Catalog through the configured workflow into the diagnostic booking/operational record.
NFR-4	Performance	Normal test-request submission and workflow processing should avoid unnecessary manual steps and excessive client-side processing; server-side orchestration should remain maintainable.
NFR-5	Availability	The solution is expected to be accessible whenever the deployed ServiceNow instance and required platform services are available; no independent uptime guarantee is asserted by this project.
NFR-6	Scalability	The design should allow additional diagnostic test categories, variables, approval rules, fulfilment paths and notification rules without redesigning the complete solution.
NFR-7	Maintainability	Prefer Service Catalog, Flow Designer, UI Policies and ACL configuration. Custom scripting should be limited to necessary behaviour and the workflow should remain understandable to another administrator.
NFR-8	Data Integrity	Values captured in the diagnostic catalog should remain consistent when mapped to the booking/operational record. Mapping and validation errors should be detectable during testing.
NFR-9	Compatibility	The requester experience should operate in the intended ServiceNow Service Portal/catalog context and use supported platform capabilities for workflow and notifications.
NFR-10	Auditability & Troubleshooting	Administrators should be able to inspect workflow execution, approval/condition paths, stored values and available execution information to diagnose failures.

Future / Out-of-Scope Requirements for the Current Version

- Additional diagnostic test categories and specialized catalog items beyond the currently demonstrated test-booking scope.
- Multi-level approval chains such as specialist approval followed by administrative approval, unless configured in the implementation.
- Automated integrations with external laboratory systems, medical devices, payment gateways or other external systems.
- Automated fulfilment task creation and closure rules beyond the workflow currently demonstrated.
- Advanced SLA calculation, escalation and performance dashboards.
- External messaging channels such as SMS, WhatsApp or other notification integrations.
- AI-based diagnostic recommendation, priority prediction or anomaly detection.
- Advanced diagnostic analytics and operational dashboards.

Requirement Traceability Summary

The critical end-to-end requirement path is: patient/requester enters valid diagnostic information → request is submitted through the Service Catalog → workflow retrieves configured variables → diagnostic booking/operational record is created with an initial Requested state → approval is requested where configured → the approval result is evaluated → the record is updated to the appropriate status → the configured notification is sent → the resulting request is available for diagnostic fulfilment, report handoff and tracking. This path defines the highest-priority testing and troubleshooting scenarios, especially variable mappings, record creation, approval routing, condition logic and notification execution.